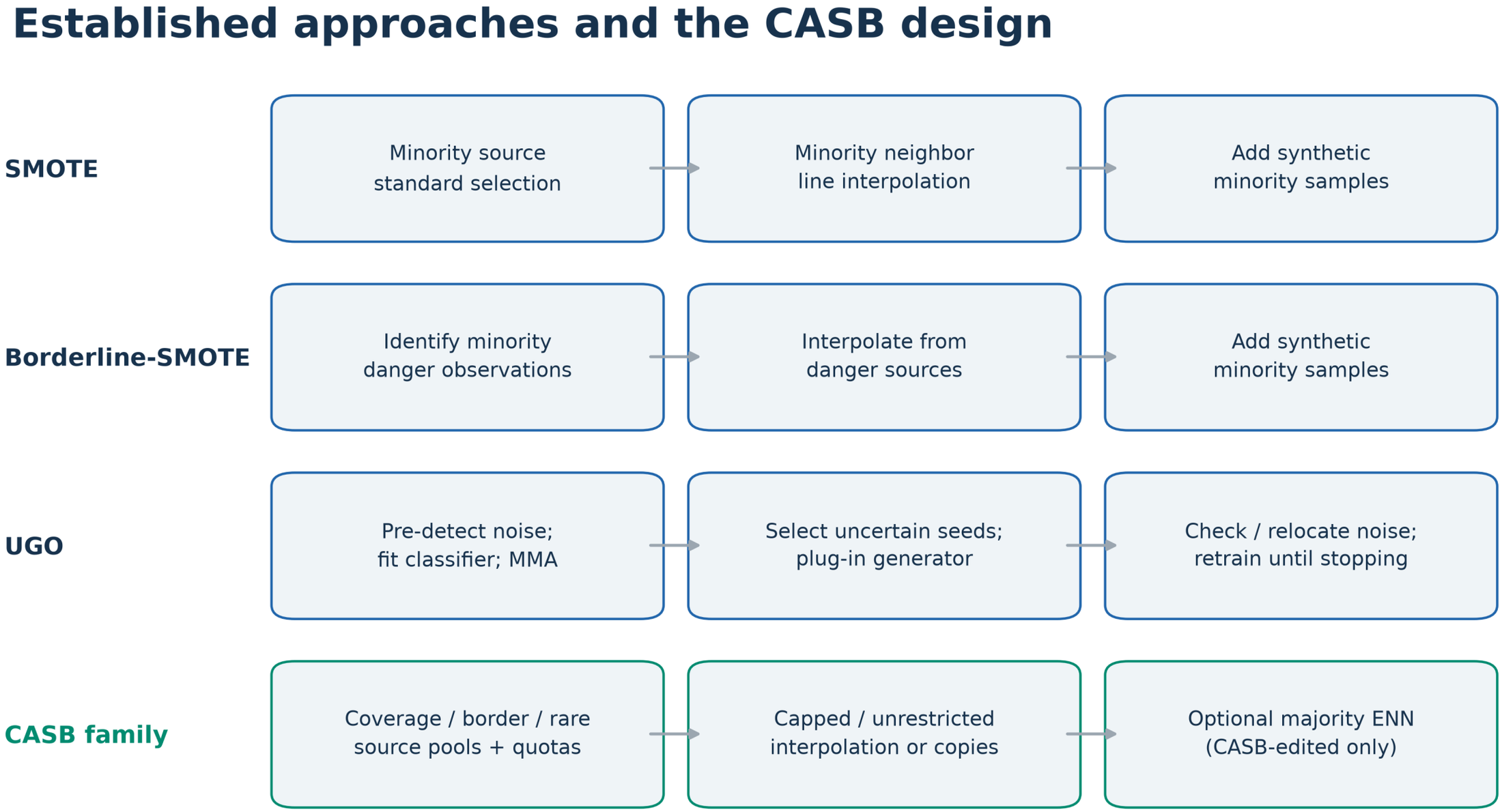
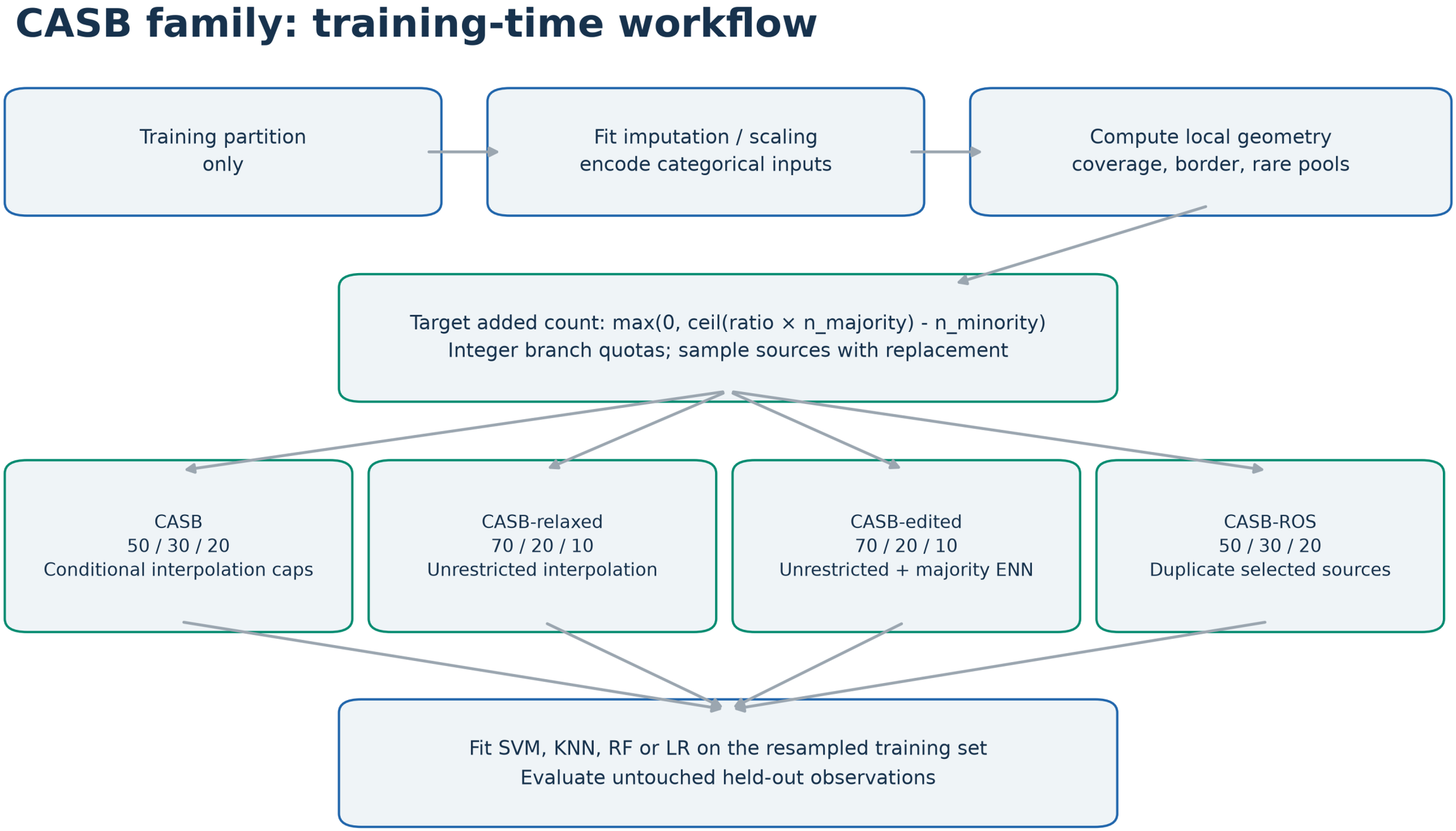
Figure 1. Prior mechanisms and CASB



CASB changes allocation and generation control. It does not implement UGO's uncertainty loop or noise relocation.

Original schematic comparing established pipelines with the current CASB family. UGO is summarized from Saglam and Cengiz (2026), Sections 3.1-3.4 and Figs. 2-4; it was not reimplemented or benchmarked here. CASB is independently defined by its neighborhood pools, quotas and generator options. These differences are descriptive, not proof of novelty. SMOTE: Chawla et al. (2002), doi:10.1613/jair.953. Borderline-SMOTE: Han et al. (2005), doi:10.1007/11538059_91. UGO: doi:10.1016/j.knosys.2026.115886.

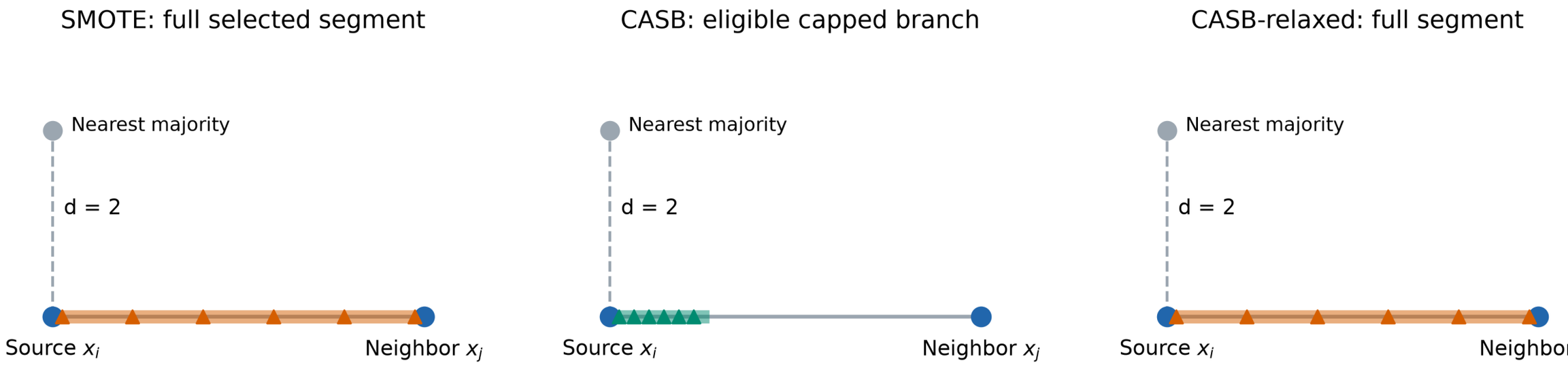
Figure 2. CASB family workflow



Workflow for the implemented variants. Percentages are quotas for newly added observations, not disjoint fractions of the original data. Pools overlap; an empty branch draws sources from coverage while retaining its branch-specific interpolation rule. Current experiments use no optional CASB majority pre-drop. The sampling ratio is selected using the existing inner-validation objective; the three ratio candidates are 0.5, 0.75 and 1.0. ENN is majority-only, three neighbors, kind_sel="mode". Preprocessing is refit within each training split.

Figure 3. Interpolation mechanics

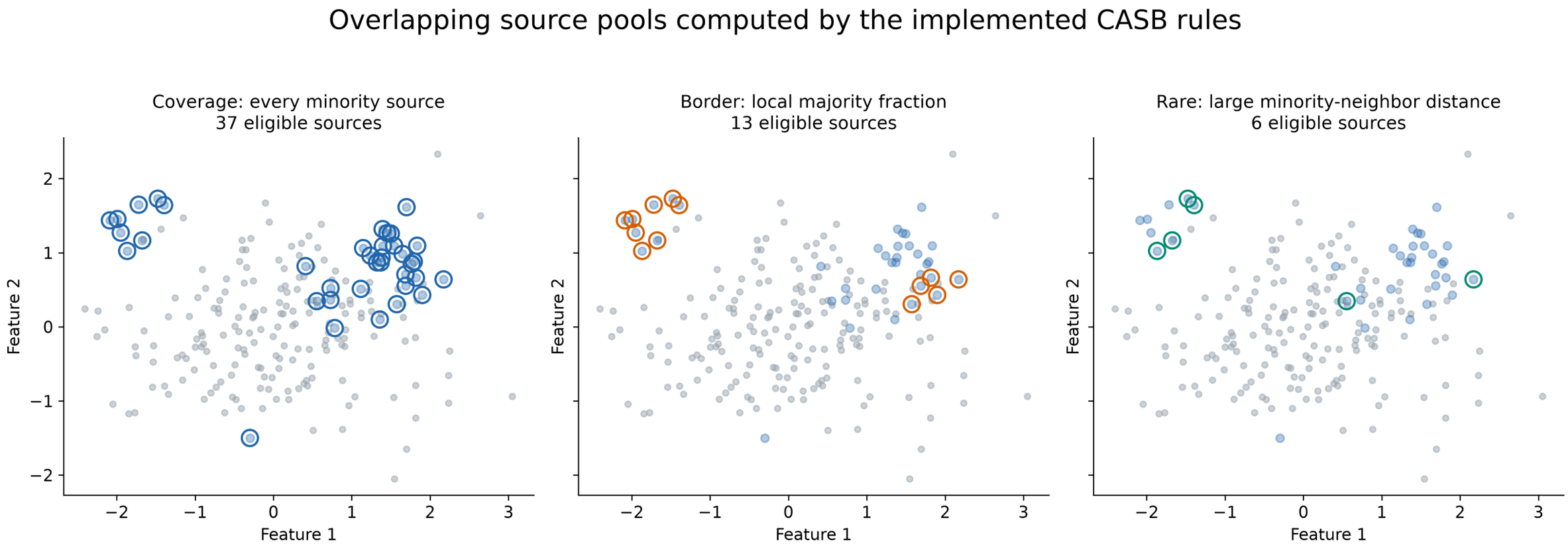
$z = x_i + \lambda(x_j - x_i)$: same interpolation form, different allowed step



Capped example: $L = 4$, $d = 2$, $c = 0.5$; $\lambda_{max} = \min(1, cd/L) = 0.25$. Rare-branch cap uses $c = 0.35$.

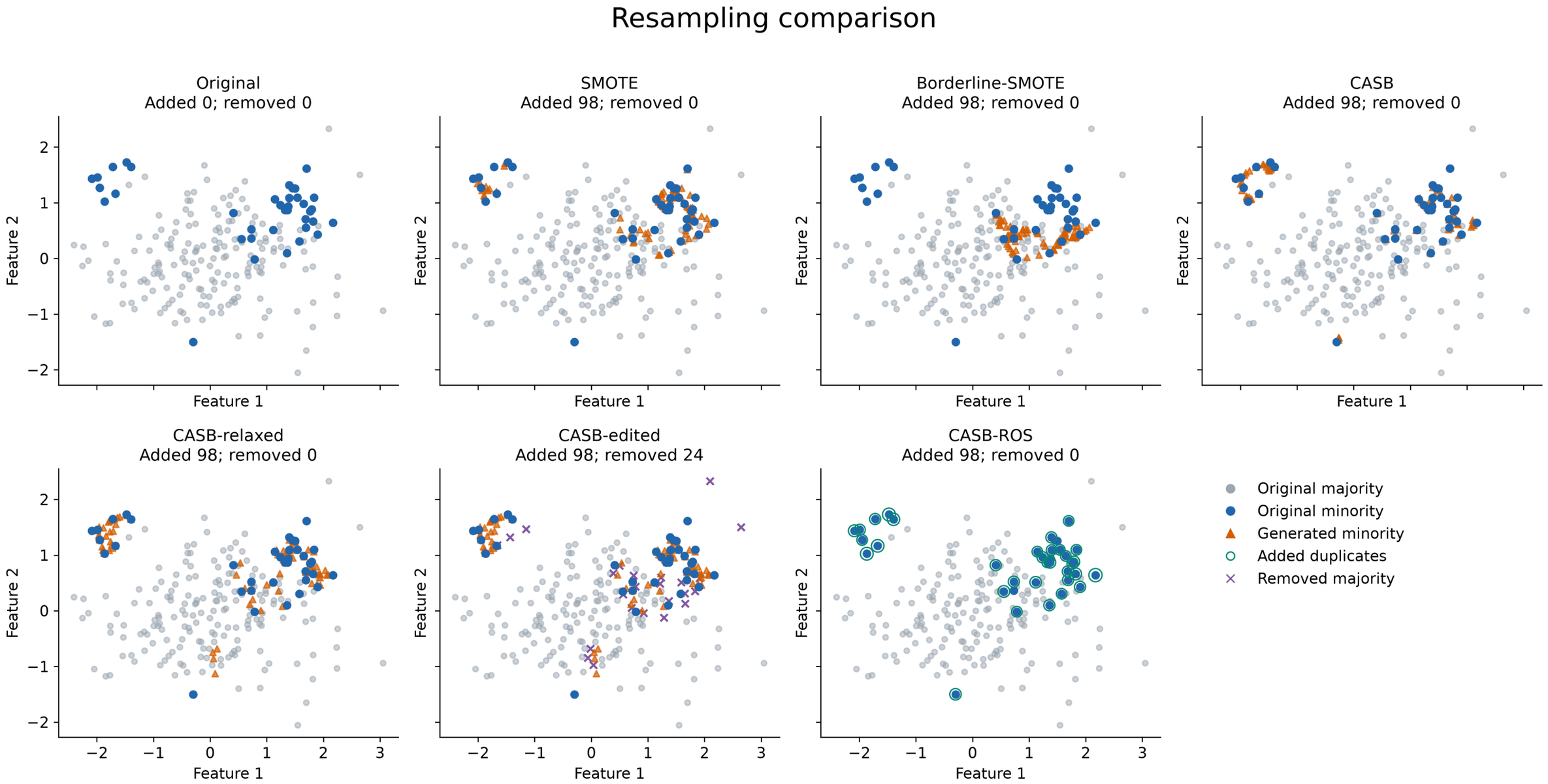
Analytical illustration, not measured data or randomly generated results. The source-neighbor pair is held fixed to isolate the interpolation rule. CASB caps non-safe coverage and border steps at min(1,0.5d/L), and rare steps at min(1,0.35d/L). Safe coverage sources with local majority fraction <=0.2 remain unrestricted. Zero-length edges use lambda=0. CASB-relaxed and CASB-edited use unrestricted interpolation; CASB-ROS sets lambda=0. Blue dots are minority observations, gray is a majority observation, triangles indicate possible synthetic positions.

Figure 4. Coverage, border and rare sources



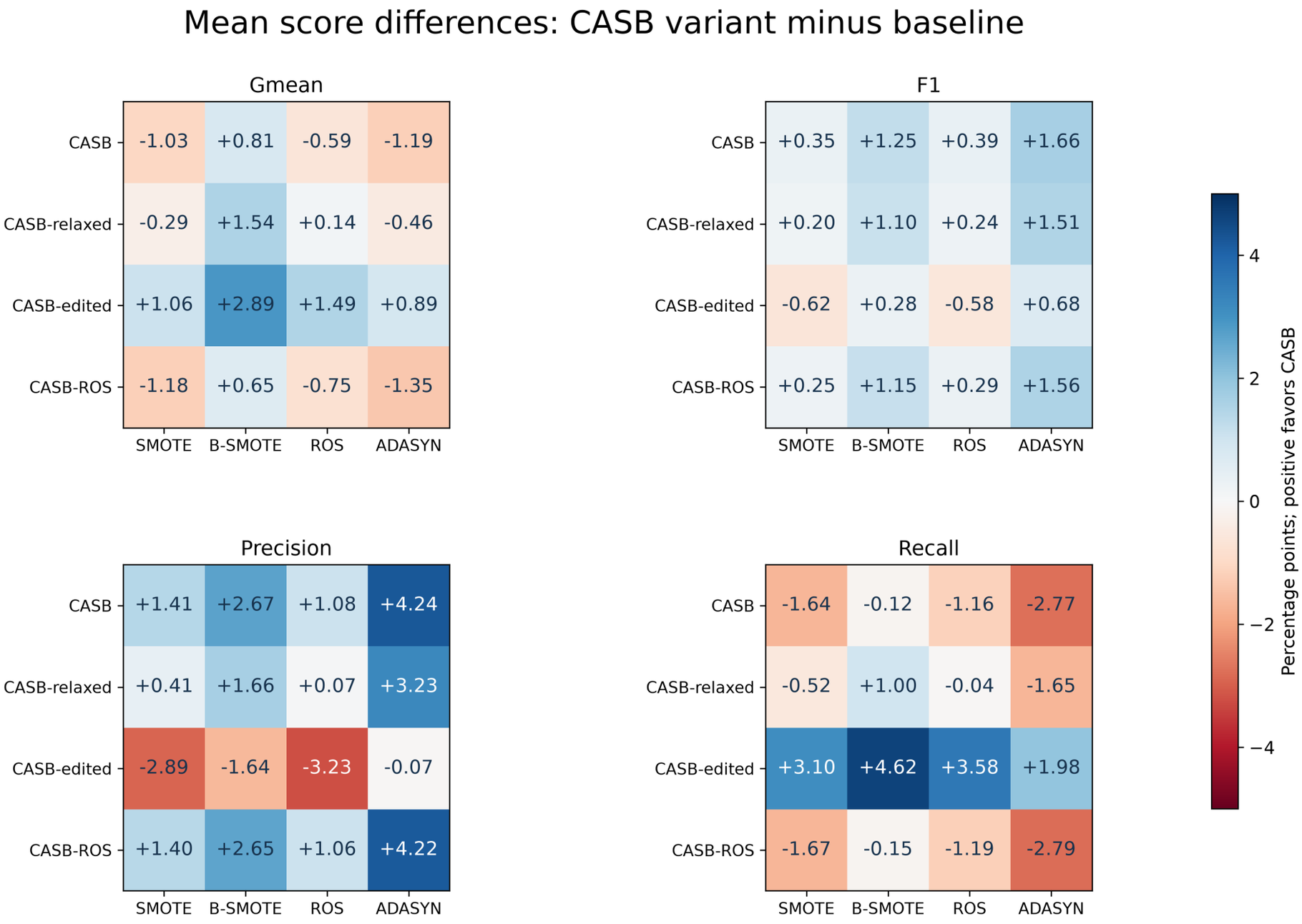
Fixed artificial two-dimensional dataset (seed 42), with pool membership computed using the implementation rules and verified against CASB diagnostics. Rings identify eligible sources, not generated observations. Coverage includes all minority sources, even potentially noisy observations. Border requires 0.2<majority fraction<=0.7; rare requires third-minority-neighbor distance >= its 75th percentile. Both require at least two minority neighbors among up to ten full-data neighbors and positive nearest-majority distance. Pool overlap is intentional; a sparse point is not automatically a verified rare subpopulation.

Figure 5. Controlled resampling example



Actual executions of the shared samplers on the artificial dataset of Figure 4. All methods use target ratio 0.75 and seed 42, isolating generation and cleaning behavior from ratio tuning. New samples are identified through sampler provenance; CASB-edited uses generator_.generated_ because ENN reorders output rows. CASB-ROS ring sizes reflect duplicate multiplicity. This is an illustrative case, not evidence of superior classification or noise handling. No UGO results are fabricated. All original minority samples are retained by CASB-edited.

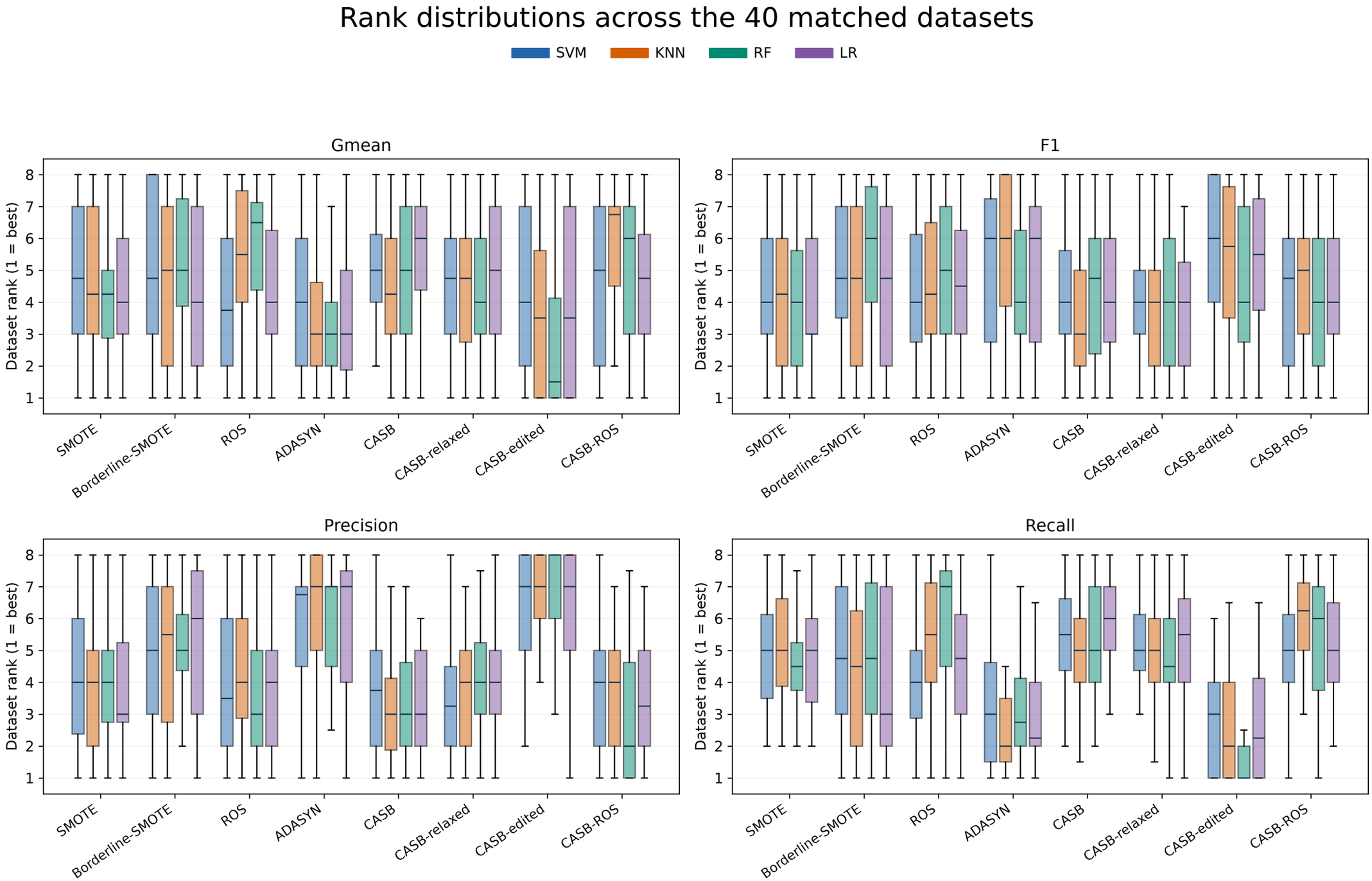
Figure 6. Average benchmark differences



40 complete datasets; equal dataset and classifier weights. Descriptive averages, not significance tests.

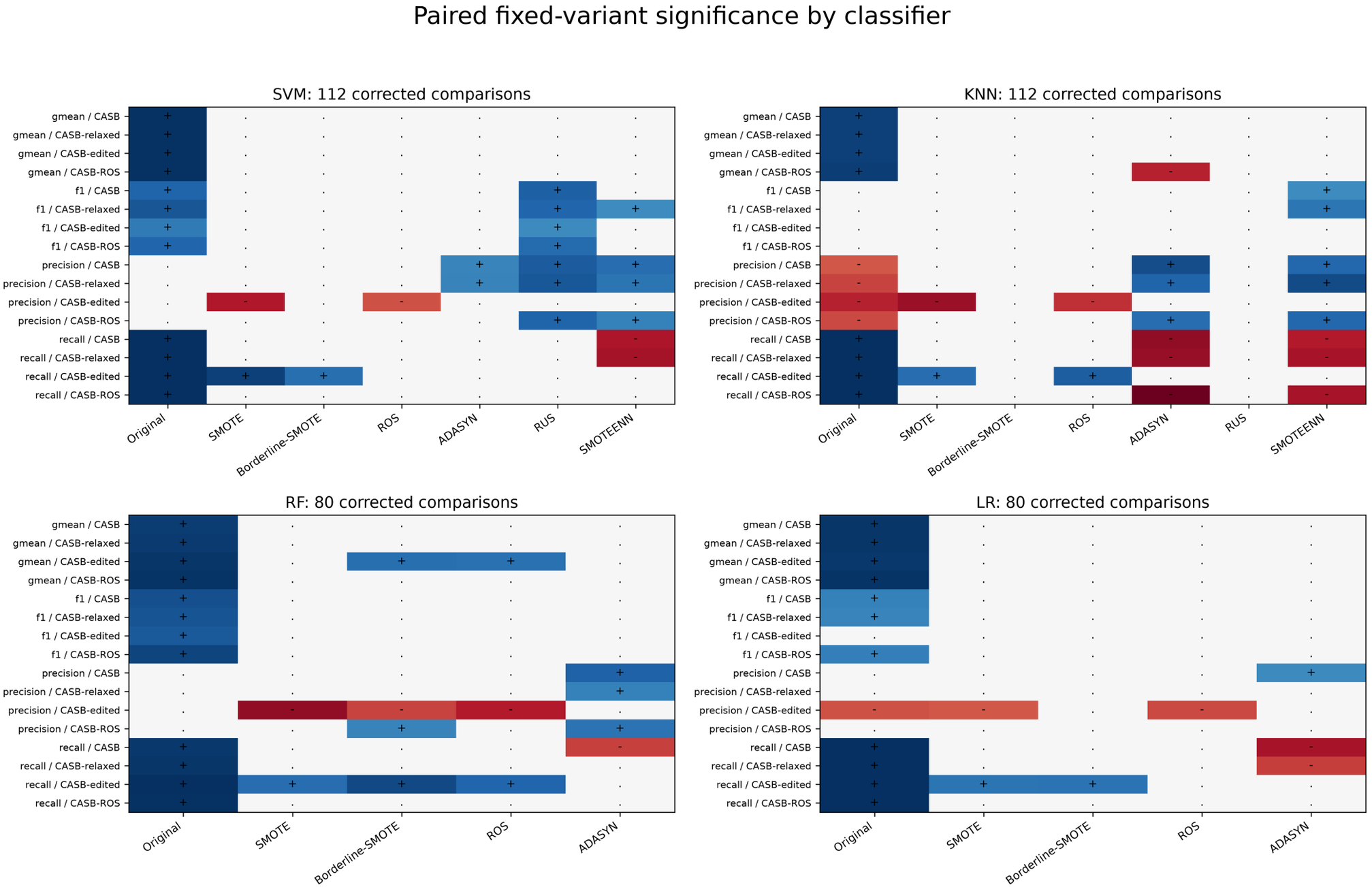
B-SMOTE denotes Borderline-SMOTE. Differences in equal-dataset-weight mean scores on 40 complete datasets. Each dataset/method/classifier first averages five supplied folds and two seeds; SVM, KNN, RF and LR then receive equal weight. Four datasets with 52 ADASYN failures are excluded from all methods in this matched comparison. No best-of-family maximum or composite metric is tested or displayed. Positive cells favor the named fixed CASB variant. These averages do not establish statistical significance.

Figure 7. Dataset-level rank distributions



Ranks of eight resampling methods, computed separately for each classifier and metric from the per-dataset fold/seed means. Lower ranks are better, tied scores receive average ranks. Boxes represent the interquartile range, lines the median, whiskers extend to 1.5 IQR; outlier markers are omitted for readability. Ranks concern 40 datasets, not 400 independent folds. Related source-dataset variants are correlated. The no-resampling reference is not included in these eight-method ranks.

Figure 8. Classifier-specific adjusted paired comparisons



\+ significant benefit; - significant harm; . not significant. Color: paired rank-biserial effect.

Two-sided Wilcoxon tests reproduced independently in statistical result. Holm correction covers 112 fixed-variant comparisons across four metrics for SVM/KNN, including no resampling, RUS and SMOTE-ENN; RF/LR use 80 comparisons including no resampling. Classifiers are kept separate. Pairwise units are 44 complete datasets, or 40 for ADASYN. Plus: significant CASB benefit; minus: significant harm; dot: non-significant at 0.05. Color encodes rank-biserial effect. Source-family sensitivity and exact sign tests are in the statistical report. No tests apply to the retrospective family maximum. Classifier retention was post hoc; findings are exploratory.